PHARMACOGENOMICS · CLINICAL PHARMACY

Locked Out

Genetics and statin response in familial hypercholesterolemia

Half the gates are missing. How a gene keeps LDL out of the liver, and how DNA shapes the drugs that let it back in.

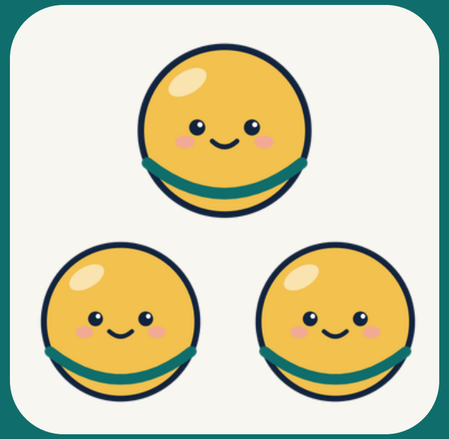


presented by Noor

PROLOGUE

What if a 10-year-old already has adult-level cholesterol?

No symptoms. No warning. Only a variant inherited from one parent, and decades of silent exposure ahead.



ROADMAP

One story, three acts



Act I · The Gene

What is broken in FH, and where



Act II · The Drug

Why statin response and safety differ between patients



Act III · The Patient

Turning a genotype into a prescription

One question runs through all three: can DNA guide lipid-lowering therapy?

PROLOGUE · CLINICAL BURDEN

The clock starts at birth



≈1 in 311

people worldwide have heterozygous FH



≈1 in 300,000

have homozygous FH (two faulty alleles)



≈3-fold

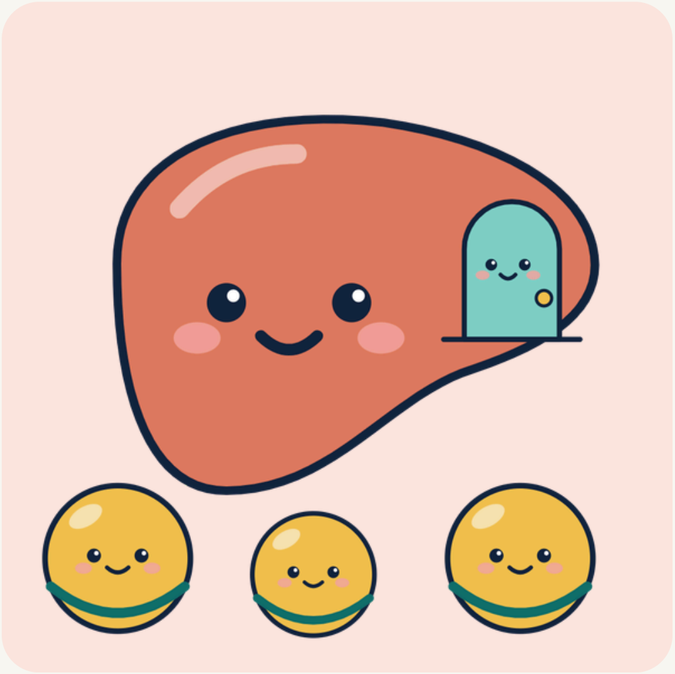
higher cardiovascular risk in variant carriers

Risk tracks **LDL-C × years**: high LDL-C from birth means disease decades earlier.

Hu et al., Circulation 2020; Cuchel et al., Eur Heart J 2023

PROLOGUE · DEFINITION

What is familial hypercholesterolemia?



Inherited: the liver cannot clear LDL from the blood

Lifelong: LDL-C is very high from childhood, above 190 mg/dL (5 mmol/L) in adults

Dangerous: premature atherosclerotic cardiovascular disease

Diagnosed by: LDL-C, family history, physical signs and a genetic test

Mach et al., Eur Heart J 2020; Methodist DeBakey Cardiovasc J 2025

ACT I

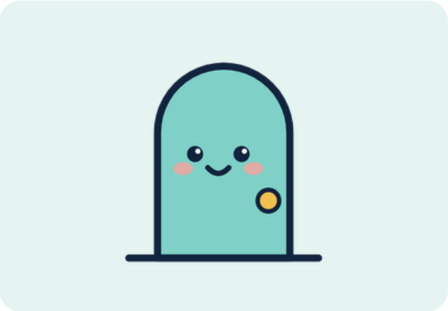
The Gene

Why can't the liver let LDL in?



ACT I · THE GENE

Four genes, one broken gate



LDLR · 85–90%

The LDL receptor itself is missing or defective



APOB · 5–10%

ApoB-100, the key on LDL, no longer fits the receptor



PCSK9 · 1–3%

Gain of function: receptors are destroyed too fast



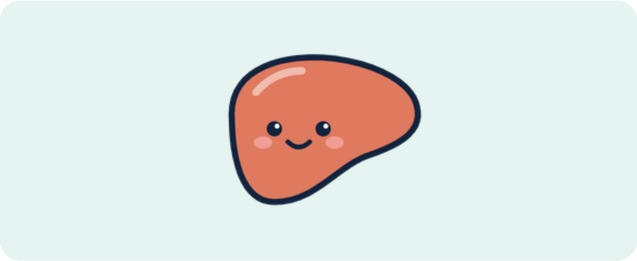
LDLRAP1 · <1%

The receptor cannot be taken into the cell (recessive)

Same end point: fewer working LDL receptors, less LDL clearance, higher LDL-C.

Cuchel et al., Eur Heart J 2023

Count the gates: one faulty allele or two



Two working alleles

No FH

●●●●●●●●●●

Full receptor activity

- Normal LDL clearance
- Statins work well



One faulty allele

Heterozygous FH

●●●●●○○○○○

About half the activity

- LDL-C above 190 mg/dL
- Statins usually work



Two faulty alleles

Homozygous FH

●○○○○○○○○○

Little or no activity

- LDL-C above 400 mg/dL
- Statins work poorly

Filled circles = working gates (schematic). Cuchel et al., Eur Heart J 2023

ACT II

The Drug

Why does the same dose work differently in different patients?



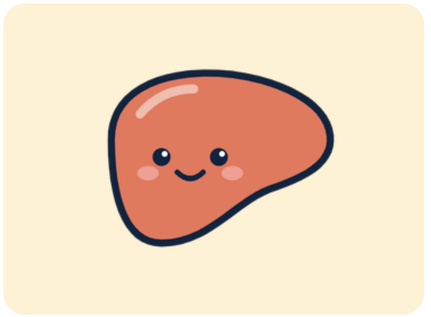
ACT II · THE DRUG

Statins make the liver build more gates



1 · Block

The statin inhibits HMG-CoA reductase



2 · Sense

Liver cholesterol falls and SREBP-2 switches on



3 · Build

More LDL receptors reach the cell surface



4 · Clear

More LDL leaves the blood: LDL-C falls by 30–50% or more

No working receptor allele, little statin effect: efficacy depends on genotype.

Mach et al., Eur Heart J 2020

ACT II · THE DRUG

Same dose, different patients



Drug-handling genes

SLCO1B1, ABCG2, CYP2C9: how much statin reaches blood and muscle



Drug-target genes

LDLR, APOB, PCSK9: how far LDL-C can fall



Non-genetic factors

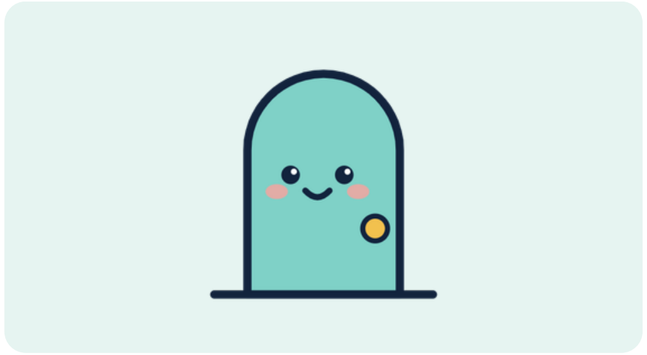
Adherence, interacting drugs, thyroid or kidney disease, diet

Handling genes mainly drive **toxicity**; target genes mainly drive **efficacy**.

Cooper-DeHoff et al., Clin Pharmacol Ther 2022

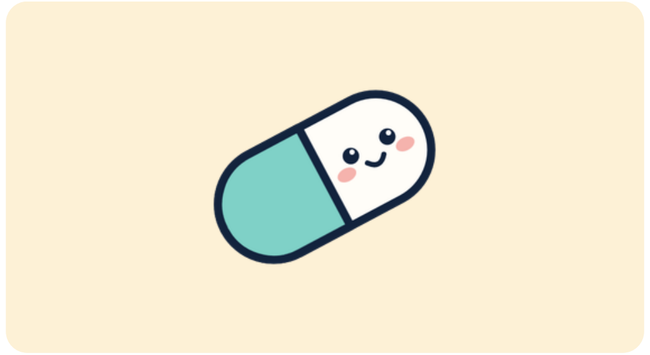
ACT II · THE DRUG

Three genes with guideline-level evidence



SLCO1B1

Liver uptake door (OATP1B1)
Simvastatin most affected
Higher myopathy risk



ABCG2

Efflux pump (BCRP)
Rosuvastatin
Higher drug exposure



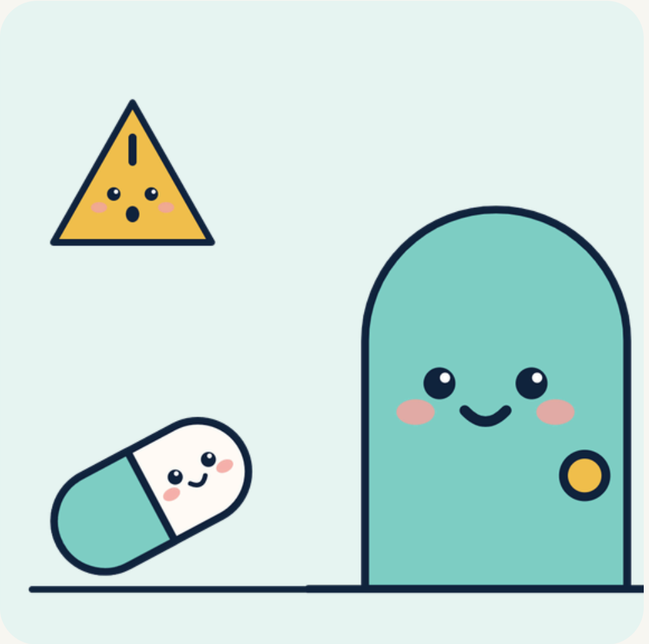
CYP2C9

Metabolizing enzyme
Fluvastatin
Slower clearance

CPIC 2022 turns each genotype into a statin- and dose-specific recommendation.

ACT II · THE DRUG

When the statin itself is locked out



SLCO1B1 c.521T>C: less statin enters the liver, so more reaches muscle.

4.5×
odds of myopathy per C allele

16.9×
odds in CC versus TT homozygotes

CPIC 2022: choose another statin, or keep simvastatin below 20 mg/day. A risk genotype is not a reason to stop statins.

SEARCH Collaborative Group, N Engl J Med 2008 (simvastatin 80 mg); CPIC 2022

BEYOND STATINS

Two ways out for LDL



Route 1 · Open more gates

Needs a working LDL receptor

Ezetimibe
NPC1L1 inhibitor: less gut absorption
≈20%

Evolocumab, alirocumab
Antibodies against circulating PCSK9
≈50–60%

Inclisiran
siRNA silencing hepatic PCSK9
≈50%

Bempedoic acid
ATP-citrate lyase inhibitor
≈15–25%

Route 2 · Take another exit

Works without the receptor

Evinacumab
ANGPTL3 antibody
≈50%

Lomitapide
MTP inhibitor
≈40–50%

Lipoprotein apheresis
Physical removal of LDL from plasma

Percentages for this route are from homozygous FH.

The patient's genotype chooses the route: when almost no gates remain, opening more of them cannot work.

Bars: approximate added LDL-C reduction in pivotal trials. Cuchel et al., Eur Heart J 2023; ESC/EAS 2025

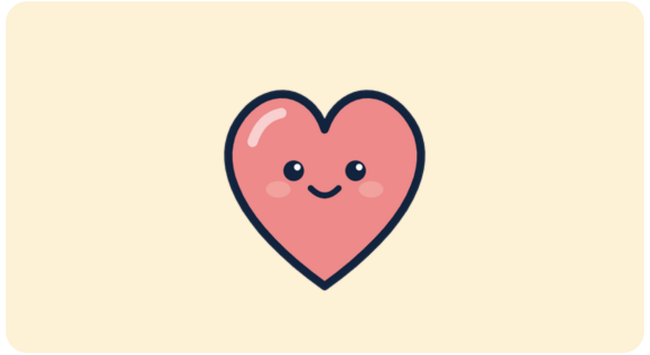
BEYOND STATINS

PCSK9: the demolition crew, switched off



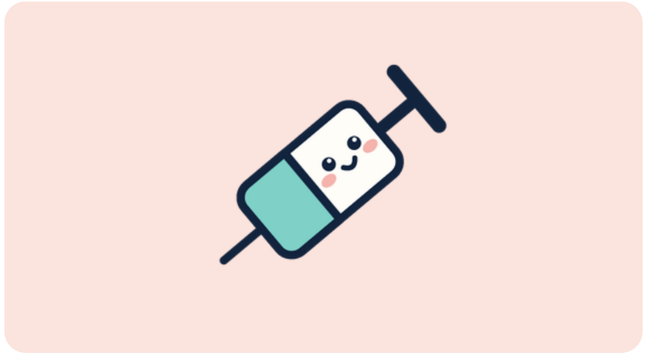
Too much PCSK9

Gain-of-function variants destroy receptors and cause FH (2003)



Too little PCSK9

Loss-of-function carriers: 28% lower LDL-C, 88% less coronary disease (2006)



Block it with a drug

Antibodies and siRNA lower LDL-C by about 50–60%

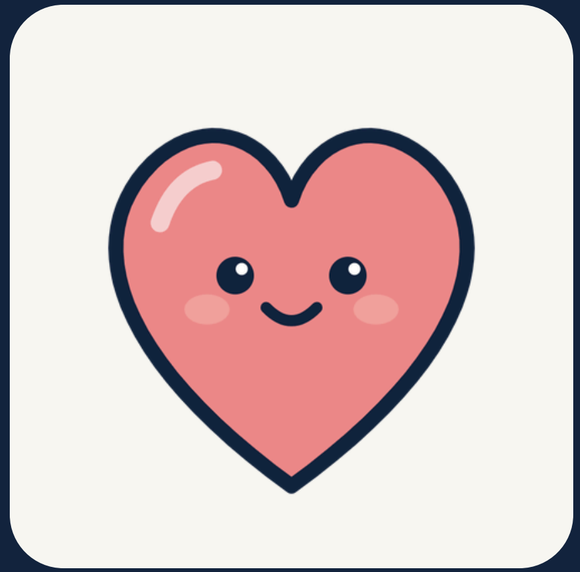
Still genotype-dependent: receptor-null homozygotes show a blunted response.

Abifadel et al., Nat Genet 2003; Cohen et al., N Engl J Med 2006; Cuchel et al., Eur Heart J 2023

ACT III

The Patient

How do we turn a genotype into a prescription?



ACT III · THE PATIENT

Four clues, one prescription



Disease genotype

Which gene, one allele or two, residual receptor activity



Pharmacogenotype

SLCO1B1, ABCG2 and CYP2C9 function



Phenotype

LDL-C, Lp(a), age, existing ASCVD



Response

LDL-C fall and tolerability at follow-up

Together they decide: **which drug, what dose, which add-on, how to monitor**.

FUTURE DIRECTIONS

Beyond statins: can diet and natural compounds help?



Bergamot

Citrus bergamia polyphenols are being studied for possible effects on cholesterol synthesis and LDL metabolism.

Where it would fit

1. Genetic FH diagnosed
2. Lifestyle and heart-healthy diet
3. Statin therapy when indicated
4. Poor response or intolerance: individualized non-statin therapy
5. **Investigational:** adjuncts such as bergamot

Do not say

"A natural replacement for statins"

Say instead

"A potential adjunct under investigation for additional LDL-C reduction"

Diet alone generally cannot overcome the genetic defect in FH.

Systematic reviews: Crit Rev Food Sci Nutr 2020; Phytother Res 2022

PROPOSED RESEARCH QUESTION

Could genetically guided treatment combined with dietary and nutraceutical interventions improve LDL-C control in young patients with familial hypercholesterolemia?



Personalized lipid management in young patients with FH: from pharmacogenomics to dietary and nutraceutical interventions

SOURCES

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